Samir Akhalil

AI & Machine Learning Engineer (M.Sc. 2027) · LLM Agents · Generative AI · Backend & APIs

Karlskrona, Sweden | Seeking an AI-assisted development thesis, spring 2027 · available from January 2027
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PROFILE

M.Sc. Engineering student in AI and Machine Learning at Blekinge Institute of Technology (graduating 2027) with hands-on experience building software around LLMs. Currently building an LLM-based incident-triage agent for Trafikverket on self-hosted models with a tool-calling agent loop. Has delivered a Flask backend with REST-style routes and SQL Server integration to Ericsson's requirements, and works with Git, Docker, Kubernetes and GitHub Actions CI. Used to designing controlled experiments and choosing metrics that reflect the real decision rather than a single headline number. Fluent in Swedish, English and Arabic.

TECHNICAL SKILLS

Languages	Python, SQL, C#, JavaScript, C/C++, Java
Generative AI	LLM agents with tool calling, self-hosted LLMs (Gemma), NLP, transformers and embeddings, Hugging Face, PyTorch
Software engineering	Flask, SQLAlchemy, REST APIs, PostgreSQL, Microsoft SQL Server, MySQL
DevOps & tools	Git, GitHub Actions (CI/CD), Docker, Kubernetes, Grafana, Prometheus, Linux
Evaluation	Controlled experiments, precision, recall, F1, attack success rate, data-quality analysis

EXPERIENCE

AI Engineering Intern – Incident-Triage Agent Sep 2026 – Jan 2027
Trafikverket (Swedish Transport Administration) · via BTH

Ten-person student team building an AI agent that triages production alerts for Trafikverket's operations teams.

- Designing an agent that reads the metrics, logs and source code behind an alert and proposes a diagnosis with next steps, so on-call engineers start from a hypothesis instead of a blank page.
- Building it on self-hosted LLMs (Gemma) with a tool-calling agent loop, so no operational data leaves the organisation's network.
- Integrating with Grafana, Prometheus and Kubernetes through a Python backend.

Machine Learning Engineer – Sensor Classification 2026 – Present
Axiogreen · Sweden

Classifying building and energy sensor types from metadata and time-series measurements.

- Mapped PostgreSQL tables of sensors, devices, locations and measurement history and standardised inconsistent names, categories, units, labels, thresholds and multipliers.
- Analysed more than 7.5 million time-series records from about 130 sensors for missing values, anomalies, value ranges, sampling frequencies and data gaps, and documented the workflow in Jupyter notebooks and Markdown reports.

Full-Stack Developer – Access-Control System 2025 – 2026
Ericsson (client project via BTH) · Karlskrona, Sweden

Student team delivering an access-control system for shared Windows computers to Ericsson's requirements.

- Built the Flask administration platform (SQLAlchemy, REST-style routes) for users, access schedules, account expiry, temporary access codes and login history.
- Implemented schedule-based authorisation that checks account status, expiry, weekday and permitted time windows before granting access.
- Connected the platform and Microsoft SQL Server to a C# Windows Forms lock-screen application handling card-based login, session control and audit logging.

AI & Software Engineer – Smart-Building Analytics 2024 – 2025
Meliox AB · Sweden

Cross-functional industry project on sensor data for energy-efficient building control.

- Applied clustering and predictive modelling to find patterns in building-sensor behaviour, and contributed to the data-flow design.
- Built dashboard prototypes presenting sensor patterns, alerts and recommendations to building operators.
- Worked in agile sprints with Git-based workflows and CI/CD alongside the industry partner.

SELECTED PROJECTS

- **Data-poisoning attacks on DistilBERT (AI security)** (GitHub). Designed controlled experiments measuring how label-flipping and backdoor attacks degrade a DistilBERT sentiment classifier on IMDB (accuracy, F1, confusion matrix, attack success rate) and tested IsolationForest in embedding space as a detection defence.
- **Hospital readmission prediction** (GitHub). End-to-end pipeline predicting 30-day readmission risk: preprocessing, feature engineering, and supervised vs. semi-supervised models compared on accuracy, precision, recall and F1 (Python, pandas, scikit-learn).

EDUCATION AND LANGUAGES

M.Sc. in Engineering – AI & Machine Learning 2022 – 2027
Blekinge Institute of Technology (BTH) · Karlskrona, Sweden

Relevant coursework: Advanced Machine Learning, Applied Artificial Intelligence, Security in AI Systems, Intelligent Data Analysis, Research Methodology, Mathematical Statistics, Linear Algebra, Multivariable Calculus

Languages: Swedish (fluent) | English (fluent) | Arabic (fluent) | Turkish (basic)